MATHIQO

How Our Maths App Will Work

The whole design, explained without the boring bits

1. Maths is a skill tree (yes, like in video games)

In games you can't unlock the cool ability until you've earned the ones before it. Maths is exactly the same. We broke all of school maths into about **a thousand small skills** — not “Chapter 7: Fractions”, but tiny, precise things like “add two fractions with different denominators”. Then we connected them with arrows meaning *learn this one first*:

The app never throws you into a skill before you've unlocked the ones feeding into it. And when you're stuck, it follows the arrows *backwards* to find out exactly which earlier skill is wobbly — like tracing a chain to find the broken link.

2. Same maths everywhere on Earth

Half plus a third works the same in Mumbai, Tokyo and New York — only the wrapping changes. So we store each question's **maths** once, and swap the wrapping per country: in India, “Ravi ate 1/2 of a paratha and Meera ate 1/3…”; in America, “Jake ate 1/2 of a pizza and Emma ate 1/3…” Same question. Same difficulty. Different costume. That's how one question bank can serve kids in every country.

3. Questions come from machines, not lists

We don't write questions one by one. We build **question machines** — a question with blanks in it, plus rules for what's allowed in the blanks. One machine can print **hundreds of different questions**, so you always get fresh numbers and nobody can just memorise the answers. And before any question reaches a student, a computer *solves it first* to prove the answer key is right. A robot helps write the machines, a computer checks every single question, and real teachers review samples — trust, but verify.

4. Wrong answers are clues, not failures

This is the cleverest part. Try it yourself: **what is 1/2 + 1/3?**

A	**5/6**	*correct — nice!*
B	**2/5**	*you added the tops AND the bottoms*
C	**2/6**	*you added the tops, multiplied the bottoms*
D	**2/3**	*you kept one of the bottoms*

Every wrong option is planted on purpose. Each one matches a **specific wrong rule** that some brains invent — so your wrong answer tells the app exactly *what you were thinking*. It's like a detective: the

mistake is the fingerprint. Pick B and the app knows you need a 30-second refresher on equivalent fractions — not the whole chapter again. One slip is ignored (everyone slips); it only acts when it sees the same fingerprint a few times.

5. It levels up exactly when you do

Every skill has a difficulty dial, not just easy/medium/hard stickers. 1/2 + 1/4 is gentler than 1/2 + 1/3, which is gentler than 5/6 + 3/8 — same skill, dial turned up. The app estimates how strong you are at each skill and serves questions *just* past your comfort zone — hard enough to be interesting, never so hard it's hopeless. And here's the honest part: we don't fully trust our own difficulty guesses. If thousands of kids ace a question we labelled “hard”, the data wins and the label changes.

6. The app gets smarter every single day

Every answer from every student teaches the system something: which questions are good, which wrong options nobody ever picks (boring — replaced), and sometimes a wrong answer pattern **nobody predicted** — a brand-new “wrong rule” we've just discovered. That discovery instantly improves every future question. Most question banks just sit there. This one learns.

7. What gets built first

Not all of maths at once — that would be a great way to build ten broken things. First: **just fractions**, roughly Classes 3–6, in India plus one more country. About 30–40 skills, a few hundred question machines, every question checked by hand. Once real students have used it and the system has learned from their answers, it grows outward — decimals, geometry, algebra — using the same map, the same machines, the same detective tricks.

THE BIG IDEA

We're not building a giant pile of questions. We're building a map of how maths fits together — the questions are just the map talking to you.

*P.S. If you answered **2/5** up in section 4 — congratulations, you just demonstrated misconception MISC.FRAC.ADD-ACROSS, the single most famous wrong rule in all of fractions. You're in excellent company: researchers have been catching kids (and adults) doing it since 1981.*